

JUNWEI DENG

junweid2@illinois.edu — 217-974-6501 — theaperdeng.github.io — Google Scholar — LinkedIn

RESEARCH SUMMARY

My research builds predictive models of the training process itself: given an intervention — removing training data (*training data attribution*) or perturbing inputs — predict the resulting model without retraining. These methods power applications from copyright attribution to data selection to safe RL post-training, and are implemented in *dattri*, the open-source library for data attribution I created.

EDUCATION

University of Illinois Urbana-Champaign

Ph.D. Candidate in Information Sciences

Advisor: Prof. Jiaqi Ma

2023.8 - 2027.4 (*expected*)

Urbana, U.S.

University of Michigan

M.S. in Information

2019.9 - 2021.4

Ann Arbor, U.S.

Shanghai Jiao Tong University

B.S.E. in Electrical and Computer Engineering

2016.9 - 2020.8

Shanghai, China.

PUBLICATIONS

• *Training Data Attribution*

How Faithful Is Trajectory-Based Data Attribution? Error Sources, Remedies, and Practical Guidelines.

J. Deng, P. Hu, S. Jin, H. Lu, J. T. Wang, S. Zhang, J. Ma.

In Submission.

A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI.

J. Deng*, Y. Hu*, P. Hu*, T. Li*, S. Liu*, et al.

In Submission.

OATS: Online Data Augmentation for Time Series Foundation Models.

J. Deng, C. Xu, J. Ma, M. Jin, J. Bian.

In Submission.

Efficient Ensembles Improve Training Data Attribution.

J. Deng*, T. Li*, S. Zhang, J. Ma.

TMLR 2026.

Taming Hyperparameter Sensitivity in Data Attribution: Practical Selection Without Costly Retraining.

W. Wang, J. Deng, Y. Hu, S. Zhang, X. Jiang, R. Zhang, H. Zhao, J. Ma.

NeurIPS 2025.

Versatile Influence Function for Data Attribution with Non-Decomposable Loss.

J. Deng, W. Tang, J. Ma.

ICML 2025.

Adversarial Attacks on Data Attribution.

X. Wang, P. Hu, J. Deng, J. Ma.

ICLR 2025.

dattri: A Library for Efficient Data Attribution.

J. Deng*, T. Li*, S. Zhang, S. Liu, Y. Pan, H. Huang, X. Wang, P. Hu, X. Zhang, J. Ma.

NeurIPS 2024 (Spotlight).

Computational Copyright: Towards A Royalty Model for Music Generative AI.

J. Deng, X. Jiang, S. Zhang, S. Zhang, H. Lakkaraju, R. Gao, C. Donahue, J. Ma.

DPFM @ ICLR 2024 (Best Paper).

• *Other Publications*

Diff-MN: Diffusion Parameterized MoE-NCDE for Continuous Time Series Generation with Irregular Observations.

X. Zhang, J. Deng, C. Xu, H. Li, J. Bian.

ICML 2026.

Adversarial Attack on Graph Neural Networks as An Influence Maximization Problem.

J. Ma*, J. Deng*, Q. Mei.

WSDM 2022.

Subgroup Generalization and Fairness of Graph Neural Networks.

J. Ma*, J. Deng*, Q. Mei.

NeurIPS 2021 (Spotlight).

WORK EXPERIENCE

Google

Research Intern

2026.5 - now
New York, U.S.

- Investigating reward hacking and cheating behaviors in coding agents.

Microsoft

Research Intern

2025.5 - 2025.8
Beijing, China.

- Investigated data attribution methods to guide diffusion-based time series generation, achieving 10% MAPE reduction on OOD benchmarks for time series foundation models.
- Developed novel influence function guided diffusion-based generative models, contributing to TimeCraft framework (800+ stars).

Intel

Machine Learning Engineer

2021.5 - 2023.7
Shanghai, China.

- Designed and open-sourced ipex-llm (8K+ stars), pioneering quantization methods for LLMs with minimal accuracy degradation on local devices like PC.
- Published optimization techniques enabling 10+ enterprise clients to adopt Intel AI solutions for production deployments. [Intel doc](#); [Anyscale blog](#); [CVPR paper](#).

OPEN-SOURCE SOFTWARE

- Creator and lead maintainer of **dattri**, a PyTorch library for developing, benchmarking, and deploying data attribution algorithms (NeurIPS 2024 **Spotlight**).
- Contributor to **dattri-llm**, an efficient and flexible infrastructure for training data attribution at LLM scale.

AWARDS & SCHOLARSHIP

- **DPFM Workshop Best Paper Award (ICLR 2024)**, with oral presentation
- **Explorer Scholarship (2020)**, for outstanding students who went abroad for their graduate study provided by Shanghai Jiao Tong University.
- **Shanghai Outstanding College Graduate (2020)**, for outstanding students graduated in Shanghai.
- **National Scholarship (2018)**, the highest award provided to outstanding university students.

SKILLS

Programming & Tools:

- Languages: Python, C++, Shell, LaTeX
- ML Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers, Diffusers), scikit-learn
- Development: Git, Docker, Weights & Biases, Jupyter

COMMUNITY SERVICE

- **Reviewer:** NeurIPS 2024/2025/2026, ICLR 2025/2026, ICML 2025/2026, AISTATS 2025/2026, AAAI 2026
- **Student Organizer:** Workshop on Regulatable ML @ NeurIPS 2023/2024/2025